BCSE 0611 APPLICATION AND DATA SECURITY

Assignment Questions:

1. Describe the process and challenges of intercepting TLS traffic from a mobile device.
2. Explain the Android architecture layers and describe the role of each layer in securing the OS.
3. What is the significance of using MobSF during mobile application security testing?
4. What is Android rooting? Discuss its advantages and security risks.
5. Explain how Burp Suite can be used to analyze mobile application network traffic.
6. Describe all vulnerabilities from the OWASP Mobile Top 10 and provide real-world examples where possible.
7. Compare static analysis and dynamic analysis in Android malware investigation.
8. What are the major challenges in reverse-engineering obfuscated Android applications?
9. What is a Remote Access Trojan (RAT) in the context of mobile devices? Discuss countermeasures to mitigate RAT infections.

10. Define cloud computing and explain how it differs from traditional on-premises IT infrastructure.
11. Compare IaaS, PaaS, and SaaS with suitable examples for each model.
12. What are S3 bucket misconfigurations? Explain how tools like s3scanner and s3Inspector assist in identifying them.
13. Explain the characteristics of public, private, and hybrid cloud deployment models.
14. Describe the main security concerns associated with Docker containers and how they can be mitigated.
15. What is the importance of Governance, Risk Management, and Compliance in an organization's security posture?
16. Define internal and external audits. How do they contribute to information security assurance?
17. Compare the key objectives of ISO 27001 and COBIT frameworks.
18. Describe common risk mitigation strategies and explain when each strategy is most appropriate.